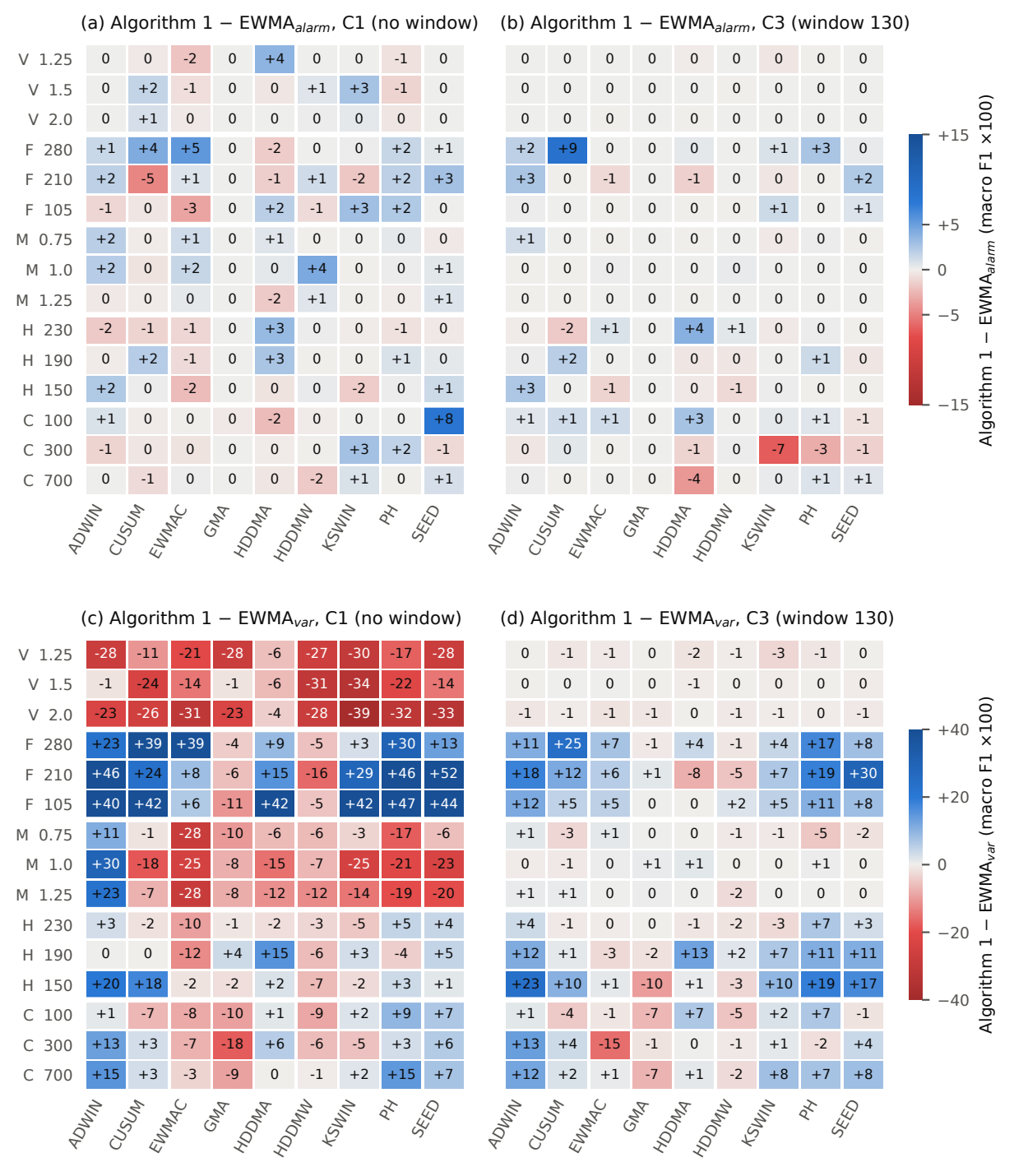
(a) Algorithm 1 − EWMA$_{alarm}$, C1 (no window)
(b) Algorithm 1 − EWMA$_{alarm}$, C3 (window 130)
(c) Algorithm 1 − EWMA$_{var}$, C1 (no window)
(d) Algorithm 1 − EWMA$_{var}$, C3 (window 130)
V 1.25 0 0 -2 0 +4 0 0 -1 0
V 1.5 0 +2 -1 0 0 +1 +3 -1 0
V 2.0 0 +1 0 0 0 0 0 0 0
F 280 +1 +4 +5 0 -2 0 0 +2 +1
F 210 +2 -5 +1 0 -1 +1 -2 +2 +3
F 105 -1 0 -3 0 +2 -1 +3 +2 0
M 0.75 +2 0 +1 0 +1 0 0 0 0
M 1.0 +2 0 +2 0 0 +4 0 0 +1
M 1.25 0 0 0 0 -2 +1 0 0 +1
H 230 -2 -1 -1 0 +3 0 0 -1 0
H 190 0 +2 -1 0 +3 0 0 +1 0
H 150 +2 0 -2 0 0 0 -2 0 +1
C 100 +1 0 0 0 -2 0 0 0 +8
C 300 -1 0 0 0 0 0 +3 +2 -1
C 700 0 -1 0 0 0 -2 +1 0 +1
V 1.25 0 0 0 0 0 0 0 0 0
V 1.5 0 0 0 0 0 0 0 0 0
V 2.0 0 0 0 0 0 0 0 0 0
F 280 +2 +9 0 0 0 0 +1 +3 0
F 210 +3 0 -1 0 -1 0 0 0 +2
F 105 0 0 0 0 0 0 +1 0 +1
M 0.75 +1 0 0 0 0 0 0 0 0
M 1.0 0 0 0 0 0 0 0 0 0
M 1.25 0 0 0 0 0 0 0 0 0
H 230 0 -2 +1 0 +4 +1 0 0 0
H 190 0 +2 0 0 0 0 0 +1 0
H 150 +3 0 -1 0 0 -1 0 0 0
C 100 +1 +1 +1 0 +3 0 0 +1 -1
C 300 0 0 0 0 -1 0 -7 -3 -1
C 700 0 0 0 0 -4 0 0 +1 +1
V 1.25 -28 -11 -21 -28 -6 -27 -30 -17 -28
V 1.5 -1 -24 -14 -1 -6 -31 -34 -22 -14
V 2.0 -23 -26 -31 -23 -4 -28 -39 -32 -33
F 280 +23 +39 +39 -4 +9 -5 +3 +30 +13
F 210 +46 +24 +8 -6 +15 -16 +29 +46 +52
F 105 +40 +42 +6 -11 +42 -5 +42 +47 +44
M 0.75 +11 -1 -28 -10 -6 -6 -3 -17 -6
M 1.0 +30 -18 -25 -8 -15 -7 -25 -21 -23
M 1.25 +23 -7 -28 -8 -12 -12 -14 -19 -20
H 230 +3 -2 -10 -1 -2 -3 -5 +5 +4
H 190 0 0 -12 +4 +15 -6 +3 -4 +5
H 150 +20 +18 -2 -2 +2 -7 -2 +3 +1
C 100 +1 -7 -8 -10 +1 -9 +2 +9 +7
C 300 +13 +3 -7 -18 +6 -6 -5 +3 +6
C 700 +15 +3 -3 -9 0 -1 +2 +15 +7
0 -1 -1 0 -2 -1 -3 -1 0
0 0 0 0 -1 0 0 0 0
-1 -1 -1 -1 0 -1 -1 0 -1
+11 +25 +7 -1 +4 -1 +4 +17 +8
+18 +12 +6 +1 -8 -5 +7 +19 +30
+12 +5 +5 0 0 +2 +5 +11 +8
+1 -3 +1 0 0 -1 -1 -5 -2
0 -1 0 +1 +1 0 0 +1 0
+1 +1 0 0 0 -2 0 0 0
+4 -1 0 0 -1 -2 -3 +7 +3
+12 +1 -3 -2 +13 +2 +7 +11 +11
+23 +10 +1 -10 +1 -3 +10 +19 +17
+1 -4 -1 -7 +7 -5 +2 +7 -1
+13 +4 -15 -1 0 -1 +1 -2 +4
+12 +2 +1 -7 +1 -2 +8 +7 +8
ADWIN
CUSUM
EWMAC
GMA
HDDMA
HDDMW
KSWIN
PH
SEED
+15
+5
0
−5
−15
Algorithm 1 − EWMA$_{alarm}$ (macro F1 ×100)
+40
+20
0
−20
−40
Algorithm 1 − EWMA$_{var}$ (macro F1 ×100)